



TRANBY COLLEGE

Psychology

Year 11 ATAR

Test 1 | Research Methods Test

Question/Answer Booklet

Student Name _____ :

Group: ☐ Friday p3/4

☐ Friday p5/6

Please tick your group.

TIME ALLOWED FOR THIS PAPER

Working time for paper:

30 minutes

MATERIAL REQUIRED/RECOMMENDED FOR THIS PAPER TO BE PROVIDED BY THE SUPERVISOR

This Question/Answer Booklet

TO BE PROVIDED BY THE CANDIDATE

Standard Items: Pens, pencils, eraser and ruler.

Special Items: Calculators satisfying the conditions set by the Schools Curriculum and Standards Authority (SCSA)—no graphics calculators permitted.

INSTRUCTIONS TO CANDIDATES

No other items may be taken into the assessment.

It is your responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you hand it to the supervisor BEFORE reading any further.

Structure of this paper

Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Number of Marks Available	Student Score
3	3	30	20	

Instructions to candidates

1. The rules for the conduct of this and the Western Australian external examinations are detailed in the *Year 12 Information Handbook 2014* and *Tranby College's Assessment Policy*. Sitting this test implies that you agree to abide by these rules.
2. This test has three (3) questions. Answer all questions. Write your answers in the spaces provided.
3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Question 1**(9 marks)**

Researchers wanted to determine the effect of drinking caffeine on students' heart rates. The researchers collected 100 Year 12 student volunteers. Fifty students were randomly assigned to Group One and fifty were assigned to Group Two.

The following sequence of tasks was undertaken by each of the participants.

- Sit calmly in a quiet room for five minutes
- Drink one cup of hot coffee (for Group One, no caffeine; for Group Two, caffeine).
- Have researcher measure heart rate.
- Return and repeat the procedure on two more consecutive days.

At the conclusion of the study, the following statistics were calculated.

Group One – no caffeine. Mean heart rate = 70 beats per minute.

Group Two – caffeine. Mean heart rate = 85 beats per minute.

(a) State the hypothesis the researchers would be testing in this study. (1 mark)

(b)

i. Identify the independent variable. (1 mark)

ii. Identify the dependent variable. (1 mark)

(c) In the above experiment, explain why the experiment is:

i. Reliable (1 mark)

ii. Valid (1 mark)

(d) Identify one source of error in the design of this experiment and suggest a way of reducing this error. (2 marks)

(e) Identify:

- i. One controllable variable (1 mark)

- ii. One uncontrollable variable (1 mark)

Question 2**(4 marks)**

A sample of secondary school students completed a quantitative self-report measure to determine their attitudes toward wearing school uniforms. The age of participants and the mean of the group, indicating their level of support for wearing a school uniform, are shown in the table below. (Higher scores indicate a stronger level of support.)

Table 1.

The Age and Score of Attitudes of Participants' Support for Wearing a School Uniform

Age	Score
14	8
15	12
16	18
17	20

- (a) State one (1) conclusion that could be drawn from this study. (1 mark)

- (b) Describe, using an example, the difference between 'population' and 'sample' when conducting research. (2 marks)

- (c) Calculate the *mean score* for these four participants. (1 mark)

Question 3**(7 marks)**

A study was conducted with a group of Psychology students to determine the relationship between the number of hours spent studying Psychology per week and the final examination score. The results of this study are shown in the table below.

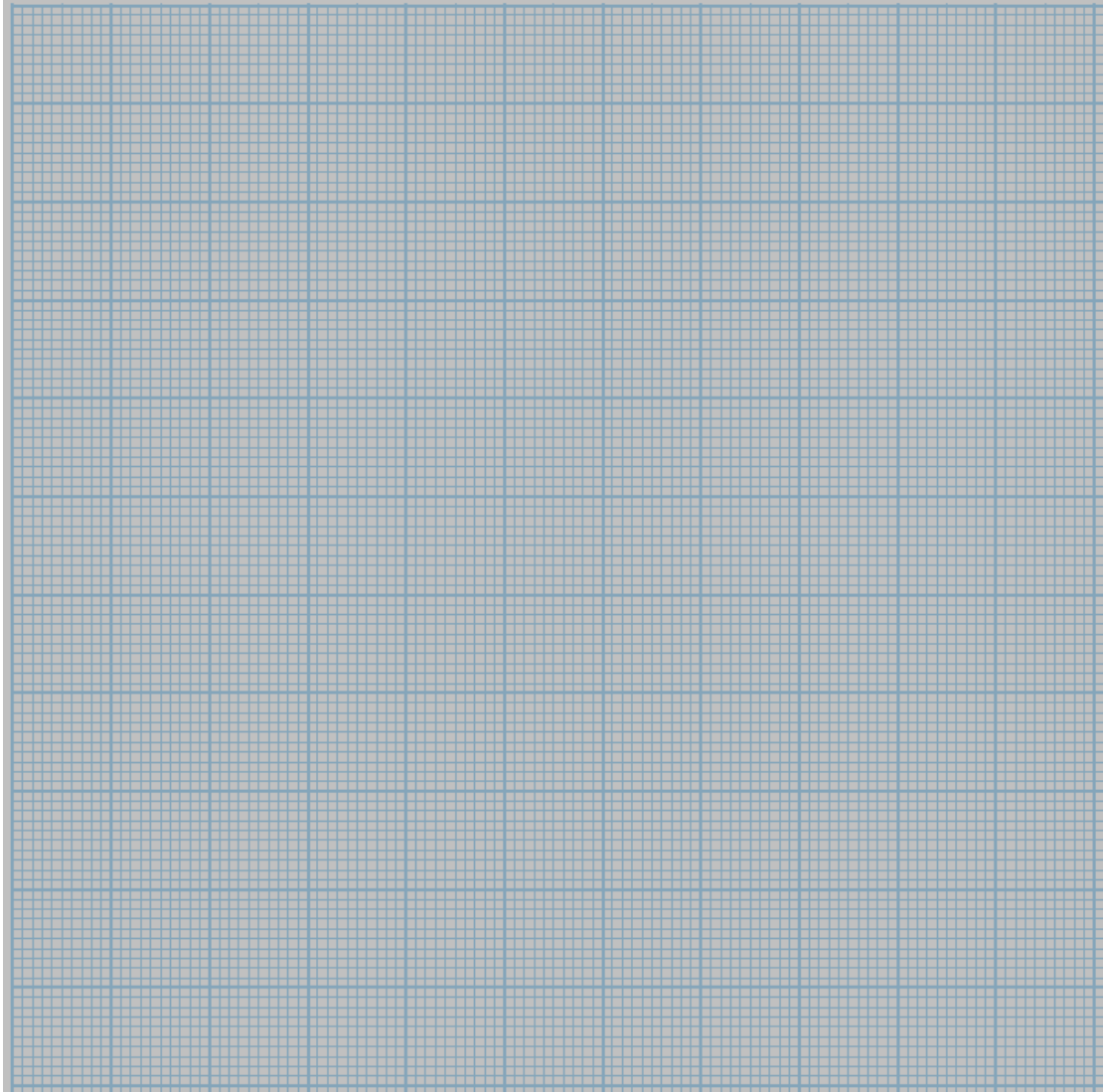
Table 2.

The Amount of Hours Studies for the Final Psychology Examination and the Respective Scores

Student	Amount of study per week (hours)	Final examination score (%)
1	13	30
2	20	70
3	4	36
4	10	65
5	13	79

- (a) State the **median** examination score from this group. (1 marks)

- (b) Plot a scattergram of these scores on the axes provided. Label the scattergram appropriately. (4 marks)



- (c) It was hypothesised that there is a relationship between the number of hours spent studying each week and examination scores achieved. With the use of the data, explain if the results given support or disprove this hypothesis?

(2 marks)

End of Test